

DATA WAREHOUSING & MINING

CE-SEM 5

VIVA QUESTIONS



“Without data
you’re just
another person
with an opinion.”

- W. Edwards Deming,
Data Scientist



Data Warehousing & Mining

1. What is a Data Warehouse?

A Data Warehouse is a centralized repository for storing integrated data from multiple sources, used for reporting and data analysis.

2. Differentiate between Data Warehouse and Database.

Databases are used for transactional processing (OLTP), while Data Warehouses are used for analytical processing (OLAP) and support complex queries.

3. What is a Data Mart?

A Data Mart is a subset of a data warehouse, focused on a specific business line or department.

4. What is Dimensional Modeling?

Dimensional Modeling is a design technique used in data warehousing that uses facts and dimensions to organize data for better query performance.

5. Explain Star Schema.

Star Schema consists of a central fact table surrounded by dimension tables, forming a star-like structure.

6. What is Snowflake Schema?

Snowflake Schema is a normalized form of Star Schema where dimension tables are further broken down into sub-dimensions.

7. What are Fact and Dimension Tables?

Fact tables contain measurable data (metrics), and dimension tables contain descriptive data (attributes).

8. What is ETL?

ETL stands for Extract, Transform, Load – the process of extracting data from sources, transforming it, and loading it into the data warehouse.

9. Differentiate between OLTP and OLAP.

OLTP handles real-time transactional operations, while OLAP supports complex analytical queries and reporting.

10. What are OLAP Operations?

Common OLAP operations include: Slice, Dice, Drill-Down, Roll-Up, and Pivot.

11. What is Data Mining?

Data Mining is the process of discovering patterns, correlations, and anomalies in large datasets.

12. What is KDD?

KDD (Knowledge Discovery in Databases) is the overall process of extracting useful knowledge from data, which includes data mining as a step.

13. List the Data Mining Task Primitives.

They include: task-relevant data, background knowledge, interestingness measures, and representation of discovered patterns.

14. What is Data Preprocessing?

Data Preprocessing involves cleaning, transforming, and organizing raw data into a suitable format for analysis.

15. What is Data Discretization?

Data Discretization is the process of converting continuous data into discrete intervals or bins.

16. What is a Decision Tree?

A Decision Tree is a predictive model that uses a tree-like graph of decisions and their possible consequences.



17.Explain Naïve Bayes Classification.

Naïve Bayes is a probabilistic classifier based on Bayes' theorem, assuming independence between features.

18.What is Confusion Matrix?

A Confusion Matrix is a table used to evaluate the performance of a classification model, showing true positives, false positives, true negatives, and false negatives.

19.What is Cross-Validation?

Cross-Validation is a technique for assessing how a model generalizes to an independent dataset by partitioning the data into training and testing sets multiple times.

20.What is Clustering?

Clustering is an unsupervised learning technique used to group similar data points together.

21.Differentiate between K-Means and K-Medoids.

K-Means uses the mean of clusters, while K-Medoids uses actual data points (medoids) as cluster centers.

22.What is Hierarchical Clustering?

Hierarchical Clustering builds a tree of clusters by either merging (agglomerative) or splitting (divisive) clusters.

23.What is Association Rule Mining?

Association Rule Mining finds interesting relationships between variables in large databases, e.g., Market Basket Analysis.

24.What is Apriori Algorithm?

Apriori is a classic algorithm for mining frequent itemsets and generating association rules using a level-wise approach.

25. What are Support, Confidence, and Lift?

Support: Frequency of an itemset. Confidence: Likelihood that Y is bought when X is bought. Lift: Measure of how much more often X and Y occur together than expected.

26. What is FP-Growth Algorithm?

FP-Growth is an efficient algorithm for mining frequent itemsets without candidate generation, using a tree structure.

27. What is Web Mining?

Web Mining involves using data mining techniques to discover patterns from web data.

28. What is Web Content Mining?

Web Content Mining extracts useful information from the content of web pages, such as text, images, etc.

29. What is Web Structure Mining?

Web Structure Mining analyzes the hyperlink structure of the web to determine the importance of pages, e.g., PageRank.

30. What is Web Usage Mining?

Web Usage Mining analyzes user interaction data like logs to understand user behavior.

31. What is PageRank?

PageRank is an algorithm used by Google to rank web pages based on the number and quality of links to them.

32. What is a Faceless Fact Table?

A Faceless Fact Table contains no measures; it is used to record events or many-to-many relationships.

33. What is Fact Constellation Schema?

Fact Constellation Schema has multiple fact tables sharing dimension tables, also known as Galaxy Schema.

34. What is Data Reduction?

Data Reduction techniques reduce the volume of data but produce the same or similar analytical results.

35. What is Data Cleaning?

Data Cleaning involves handling missing values, noise, and inconsistencies in the data.

36. What is Data Integration?

Data Integration combines data from different sources into a coherent data store.

37. What is Holdout Method in Evaluation?

The Holdout Method splits the dataset into training and testing sets for model evaluation.

38. What is Bootstrap Method?

Bootstrap is a resampling technique that involves sampling with replacement to estimate statistics.

39. What is Agglomerative Clustering?

Agglomerative Clustering is a bottom-up approach where each data point starts as its own cluster and merges iteratively.

40. What is Divisive Clustering?

Divisive Clustering is a top-down approach where all data points start in one cluster and split iteratively.

41. What are Closed and Maximal Itemsets?

Closed Itemset: An itemset with no superset having the same support. Maximal Itemset: An itemset with no frequent superset.

42. What is Multilevel Association Rule Mining?

It mines association rules at different levels of concept hierarchy, e.g., product categories and subcategories.



43. What is Multidimensional Association Rule Mining?

It involves multiple dimensions or attributes in the rule, e.g., age, income, and location.

44. What is the Role of Metadata in Data Warehousing?

Metadata provides information about the data warehouse's structure, source, and transformations.

45. What is Incremental Data Loading?

Incremental loading updates only the changed or new data since the last load, improving efficiency.

46. What is the Difference Between Classification and Clustering?

Classification is supervised learning with labeled data; clustering is unsupervised learning without labels.

47. What is the Curse of Dimensionality?

It refers to the problem where high-dimensional data becomes sparse, making analysis difficult.

48. What is a Concept Hierarchy?

A Concept Hierarchy defines a sequence of mappings from low-level to high-level concepts, e.g., city → state → country.

49. What is the Role of WEKA in Data Mining?

WEKA is an open-source tool used for data preprocessing, classification, clustering, and visualization.

50. What is HITS Algorithm?

HITS (Hyperlink-Induced Topic Search) is a link analysis algorithm that identifies hubs and authorities in web pages.